



Predictive Power of Meteorological Conditions on Forest Fire Extent

Given a range of historical weather data, can machine learning models accurately predict the extent of damage of forest fires in Montesinho park?

Introduction

This poster investigates whether meteorological data alone can accurately predict forest fire damage in Montesinho Park by training 2 machine learning models on the Montesinho dataset.

An initial data analysis highlighted 2 key issues:

- **Non-Linearity** – Features show weak correlation with the target (R2 from Linear Regression = -36.85)
- **Multicollinearity** – Significant correlations between features, reducing individual predictive reliability.

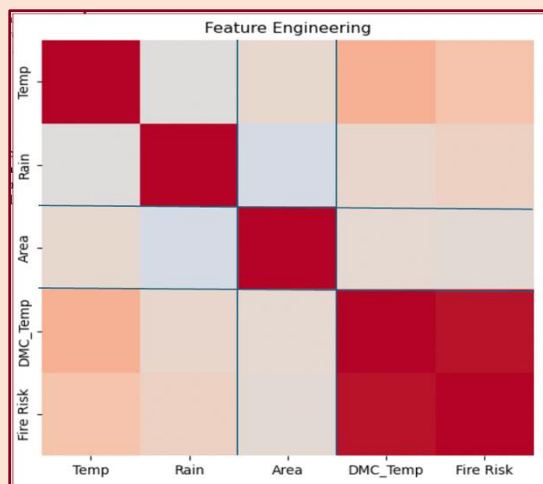
This poster presents strategies to address these challenges, compares two distinct model approaches, and draws conclusions about model performance, key findings, and limitations in relation to the research question.

Methodology

Performed feature engineering to minimise pitfall effects:

1. $y=\log(x+1)$ on area to reduce skewness, stabilise target variance.
2. Interaction features to enhance correlation with target.
3. Encoded rainfall as binary variable due to near-zero values.
4. Polynomial transformation on temp and interaction features.

These provided only a slight increase in correlation.



- A standard **80-20 split** was used due to the small dataset size. Instead of validation set, **cross validation with Grid Search** was used for hyper parameter tuning.
- **Applied Standardization** to bring features to a comparable scale, enhancing model performance.

Models selected for training:

1. Ridge Regression - Counteracts multicollinearity & overfitting
2. Gradient Boosting (XGBoost) – Counteracts non-linearity

The evaluation approach involved using metrics like R2, MAE, and RMSE, with cross-validation and hyperparameter tuning.

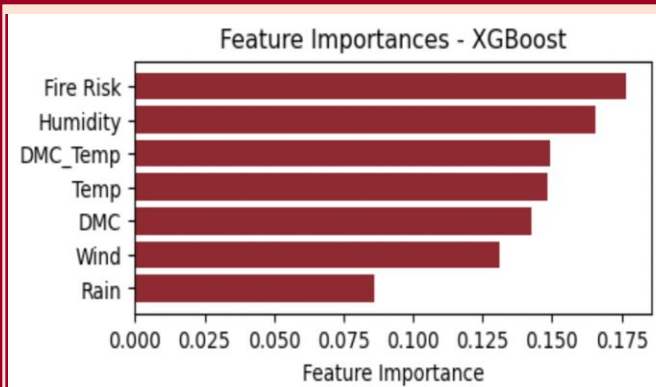
Ridge Regression – The aim with RR was to reduce multicollinearity by penalizing large coefficients using L2 regularization. Tuned α to penalize optimisation function.

$$RSS(w,b) = \sum_{i=1}^N (y_i - (wx_i + b))^2 + \alpha \sum_{j=1}^p w_j^2$$

XGBoost Regressor – Builds an ensemble of decision trees to capture complex patterns.

Hyperparameters tuned:

- **max_depth** (3) - control tree complexity
- **learning_rate** (0.01)
- **n_estimators** (100) to balance model accuracy and computational efficiency,



Other Limitations: The limited size of the dataset, impacts ability of complex models like XGBoost to generalise well. It might also lead to overfitting.

Suggestions: To better predict extent of a forest fire, consider other factors like topography, vegetation types, strength of fire response team in the area, etc.. Using a more comprehensive version of this dataset with more meteorological and time features would boost our ability to train better predictive models.

∴ These conditions alone can't predict area accurately.

	Ridge	XGBoost
MAE	1.194	1.181
RMSE	1.488	1.470
R2	0.012	0.016

Results

Both models performed similarly on this dataset

with only slight differences in error metrics. XGBoost was expected to outperform Ridge significantly. However, its very minor improvements suggest that the dataset lacks strong predictive features. The best Ridge model used $\alpha = 500$ for heavy regularization to manage multicollinearity, but even with this tuning, performance gains were minimal. XGBoost slightly improved RMSE, but neither model effectively reduced larger errors.

Conclusion

This data is NOT suitable for regression modelling.

- **Low feature importances** across the board shows no single feature has a strong influence on target.
- **Wide spread of residuals around zero** shows predictions are not closely following the pattern of the actual values. This highlights the low R2 as the model doesn't capture the variance well.

